Practice Paper C - Exam Solutions

1. (a) Students who took a test were asked to submit their marks for research purposes. 100 students responded to the call, and their marks are recorded and summarised in the grouped frequency table below.

Mark Range	10–19	20–29	30–39	40–49	50–59	60–69	70–79	80–89
Frequency	8	18	25	22	16	6	3	2

(i) Determine an estimate of the mean and standard deviation of the marks obtained. (7 marks)

Solution: To find the sum, we need to multiply the frequency of each range by the midpoint:

$$\sum x = (15 \times 8) + (25 \times 18) + (35 \times 25) + (45 \times 22) + (55 \times 16) + (65 \times 6) + (75 \times 3) + (85 \times 2)$$
$$= 4100$$

Then the mean is this divided by the total number of values (100):

$$\bar{x} = \frac{4100}{100} = 41$$

[2 marks]

The variance is given by:

$$s_x^2 = \frac{1}{n-1}\left(\sum x^2 - \frac{(\sum x)^2}{n}\right)$$

[1 mark]

We are told $n = 100$, so $n - 1 = 99$. Then, $\sum x^2$ can be found by squaring each midpoint and multiplying it by the frequency:

$$\sum x^2 = (15^2 \times 8) + (25^2 \times 18) + (35^2 \times 25) + (45^2 \times 22) + (55^2 \times 16) + (65^2 \times 6) + (75^2 \times 3) + (85^2 \times 2)$$
$$= 193,300$$

[1 mark]

We know from earlier that $\sum x = 4100$. Then

$$s_x^2 = \frac{1}{99}\left(193300 - \frac{4100^2}{100}\right) = 254.54$$

So

$$s_x = \sqrt{254.54} = 15.95$$

[2 marks]

(ii) Explain why, given this data, you can only estimate the mean, and not calculate it precisely. (2 marks)

Solution: The data has been grouped into bins, which means we can only estimate the mean using the central value of each bin - if the actual data is skewed towards one end of a bin, the true mean will be different. [2 marks]

(b) Considering the source of this data, how might you expect it to be biased? (2 marks)

Solution: Since the students volunteered to submit their marks, they might be reluctant to do this if they scored poorly on the test, so we might see a lower proportion of low-scoring test results in the sample. Other valid sources of bias also accepted. [2 marks]

(c) The actual distribution of marks for this cohort of 500 students is given below. Perform a statistical test at the 5% significance level to determine how good the fit is for this data. State your hypotheses and conclusion.

Mark Range	10–19	20–29	30–39	40–49	50–59	60–69	70–79	80–89
Frequency	40	95	125	110	90	26	10	4

(15 marks)

Solution: Performing a χ^2 test for goodness of fit. There are 8 categories, so $\nu = 8 - 1 = 7$. [2 marks]

H_0: the distribution of marks in the sample is a good fit for the data generally
H_1: the distribution is very different from the whole dataset [2 marks]

Mark Range	10–19	20–29	30–39	40–49	50–59
Obs. count	8	18	25	22	16
Exp. (/500)	40	95	125	110	90
Exp. (/100)	$\frac{40}{5} = 8$	$\frac{95}{5} = 19$	$\frac{125}{5} = 25$	$\frac{110}{5} = 22$	$\frac{90}{5} = 18$
$\frac{(O-E)^2}{E}$	$\frac{0^2}{8} = 0$	$\frac{-1^2}{19} = 0.052$	$\frac{0^2}{25} = 0$	$\frac{0^2}{22} = 0.3$	$\frac{-2^2}{18} = 0.22$

Mark Range	60–69	70–79	80–89
Obs. count	6	3	2
Exp. (/500)	26	10	4
Exp. (/100)	$\frac{26}{5} = 5.2$	$\frac{10}{5} = 2$	$\frac{4}{5} = 0.8$
$\frac{(O-E)^2}{E}$	$\frac{0.8^2}{5.2} = 0.12$	$\frac{1^2}{2} = 0.5$	$\frac{1.2^2}{0.8} = 1.12$

[8 marks]

The test statistic is then

$$\chi^2 = 0 + 0.052 + 0 + 0.3 + 0.22 + 0.12 + 0.5 + 1.12 = 2.312$$

[1 mark]

For $\nu = 7$ and 5% significance, the critical value is 14.067, and since $2.312 < 14.067$ we can accept the null hypothesis that this sample is representative of the whole cohort. [2 marks]

Total marks for question 1: [26]

2. (a) Manchester City and Manchester United play derby matches against each other roughly once a year, and of the 198 matches played so far, Manchester United has won 81. Assume the outcome of successive derbies are independent.

(i) Let U represent the number of United wins over the next 40 derby matches. Use the historical data to write down a model U_B that represents U using a binomial distribution. State the requirements for a binomial distribution and explain why this is a suitable model. (7 marks)

Solution: In order to model a situation using a binomial distribution, we need four conditions to be satisfied:

- Two outcomes - here, these are 'Manchester United wins' and 'Manchester United doesn't win' (including draws)
- Independence - we assume the outcome of each derby is independent
- Fixed number of trials - we are considering the next 40 matches
- Fixed probability of success - we can use the historical proportion of wins (41%) as a probability of success for each match

[4 marks]

Therefore this is a suitable situation to model using a binomial distribution, which we could write as $U_B \sim B(40, 0.41)$. [3 marks]

(ii) Write down an approximation to U using a normal distribution. (4 marks)

Solution: For U_B, $n = 40$ and $p = 0.41$ so we can write $U_N \sim N(np, np(1-p)) = N(16.4, 9.676)$ [4 marks]

(iii) Give one reason why this approximation is valid. (2 marks)

Solution: The sample size is $40 > 20$, so we can approximate the binomial using a normal distribution. [2 marks]

(iv) Calculate the probability that Manchester United will win more than half of the next 40 derby matches. (5 marks)

Solution: The next 40 derby matches are modelled as $N(16.4, 9.676)$, so to find the probability of a value greater than 20 we can calculate the z-score for 20 as:

$$z = \frac{x-\mu}{\sigma} = \frac{20-16.4}{9.676} = 0.532$$

[3 marks]

This is the area to the left of the line at 20, so the probability of more than 20 wins is given by:

$$1 - 0.532 = 0.468 = 46.8\%$$

[2 marks]

(b) The mean attendance at football matches at Old Trafford is 73,979, and is normally distributed. A sample of 7 recent Manchester derby matches taking place at Old Trafford was collected, and the mean was found to be 74,444 and the standard deviation 955.4.

(i) What type of statistical test should you use to test whether the mean attendance at derby games is different from the mean attendance in general? Justify your answer. (3 marks)

Solution: We could use a two-tailed one-sample t-test. The sample size is not large enough for a z-test, and we do not know the population standard deviation. [3 marks]

(ii) Perform this test at the 5% significance level, and explain the result, stating your hypotheses clearly. (10 marks)

Solution: Using a two-tailed t-test at the 5% level with $\nu = 7 - 1 = 6$. [2 marks]

H_0: $\mu = 73979$
H_1: $\mu \neq 73979$ [2 marks]

To calculate the test statistic:

$$t = \frac{\bar{x} - \mu}{\frac{s_x}{\sqrt{n}}} = \frac{74444 - 73979}{\frac{955.4}{\sqrt{7}}} = 1.29$$

[2 marks]

Look this value up in the table for the t-distribution for $\nu = 6$ with two tails, The value 1.29 lies between 1.134 and 1.440, meaning it corresponds to a probability between 0.2 and 0.3, which is not below the significance level of 5% or 0.05.

Alternatively, we could check the table to see that for 5% significance on a 2-tailed test with $\nu = 6$, the critical value for the test statistic is 2.447, and the critical region is $t > 2.44$ or $t < -2.447$. Since $t = 1.29$, this is not within the critical region.
[2 marks, for either approach]

This means we can accept the null hypothesis. There is not significant evidence to suggest attendance at derby matches is different from attendance at matches in general.
[2 marks]

[10 marks]

Total marks for question 2: [31]

3. (a) (i) Find the determinant of the matrix $A = \begin{pmatrix} 6 & 2 \\ -5 & -2 \end{pmatrix}$ (4 marks)

Solution:

$$\begin{aligned} det(A) &= \begin{vmatrix} 6 & 2 \\ -5 & -2 \end{vmatrix} \\ &= (6)(-2) - (2)(-5) \\ &= (-12) - (-10) \\ &= -12 + 10 = -2 \end{aligned}$$

[4 marks]

(ii) Calculate or write down the determinant of A^{-1}. (4 marks)

Solution:

$$\begin{aligned} A^{-1} &= \frac{1}{det(A)} \begin{vmatrix} -2 & -2 \\ 5 & 6 \end{vmatrix} \\ &= -\frac{1}{2} \begin{vmatrix} -2 & -2 \\ 5 & 6 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ -2\frac{1}{2} & -3 \end{vmatrix} \end{aligned}$$

Then

$$\begin{aligned} det(A^{-1}) &= (1 \times -3) - (1 \times -2\tfrac{1}{2}) \\ &= (-3) + 2\tfrac{1}{2} = -\tfrac{1}{2} \end{aligned}$$

Alternatively, we can note that $det(A^{-1}) = \frac{1}{det(A)}$, and obtain the same result. [4 marks]

(iii) Find the inverse of the matrix $B = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 1 & 4 & 4 \end{pmatrix}$ (10 marks)

Solution: Using Gauss-Jordan elimination:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 2 & -1 & 3 & 0 & 1 & 0 \\ 1 & 4 & 4 & 0 & 0 & 1 \end{array}\right)$$

$$\begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \end{array} \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & -2 & 1 & 0 \\ 0 & 4 & 2 & -1 & 0 & 1 \end{array}\right)$$

$$-R_2 \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 4 & 2 & -1 & 0 & 1 \end{array}\right)$$

$$R_3 - 4R_2 \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 0 & -2 & -9 & 4 & 1 \end{array}\right)$$

$$-\frac{1}{2}R_3 \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 0 & 1 & \frac{9}{2} & -2 & -\frac{1}{2} \end{array}\right)$$

$$\begin{array}{l} R_1 - 2R_3 \\ R_2 - R_3 \end{array} \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -8 & 4 & 1 \\ 0 & 1 & 0 & -\frac{5}{2} & 1 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{9}{2} & -2 & -\frac{1}{2} \end{array}\right)$$

Therefore the inverse of B is

$$B^{-1} = \begin{pmatrix} -8 & 4 & 1 \\ -\frac{5}{2} & 1 & \frac{1}{2} \\ \frac{9}{2} & -2 & -\frac{1}{2} \end{pmatrix}$$

[10 marks; lose 1 for each mistake]

(b) Here is the covariance matrix for three variables, X, Y and Z:

$$\begin{pmatrix} 60 & 32 & -4 \\ 32 & 30 & 0 \\ -4 & 0 & 80 \end{pmatrix}$$

Comment on the entries in this matrix - state what they represent, and what this tells you about the variables. (5 marks)

Solution: The diagonal elements 60, 30, and 80 indicate the variance in data sets X, Y, and Z respectively. Y shows the lowest variance whereas Z displays the highest variance.

[2 marks]

The covariance for X and Y is 32. As this is a positive number it means that when X increases (or decreases) Y also increases (or decreases). [1 mark]

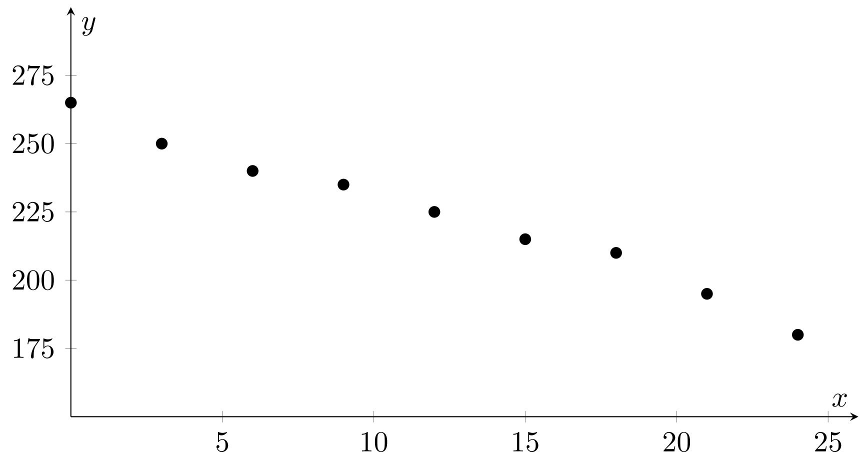
The covariance for X and Z is -4. As it is a negative number it implies that when X increases, Z decreases, and vice-versa. [1 mark]

The covariance for Y and Z is 0. This means that there is no predictable relationship between the two data sets. [1 mark]

Total marks for question 3: [23]

4. (a) A new car tyre is fitted to a wheel. The tyre is inflated to its recommended pressure of 265 kPa and the wheel is left unused. At 3-month intervals, the tyre pressure is measured:

Time after fitting (x months)	0	3	6	9	12	15	18	21	24
Tyre Pressure (y kPa)	265	250	240	235	225	215	210	195	180



(i) For this data,

$$\sum(x-\bar{x})(y-\bar{y}) = -1755$$

Calculate the equation of the least squares regression line of y on x. (10 marks)

Solution: The mean for x can be calculated to be $\bar{x} = 11.625$. [1 mark]

Then

$$b = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2}$$

$$= \frac{1755}{(0-11.62)^2+(3-11.62)^2+(6-11.62)^2+(9-11.62)^2+(12-11.62)^2+(15-11.62)^2+(18-11.62)^2+(21-11.62)^2+(24-11.62)^2}$$

$$= \frac{1755}{135.14+74.39+31.64+6.89+0.14+11.39+40.64+87.89+153.14}$$

$$= \frac{-1755}{541.26} = -3.24$$

[6 marks]

And

$$a = \bar{y} - b\bar{x} = 223.89 - (-3.24 \times 11.625) = 261.55$$

[3 marks]

(ii) Based on this equation, and confirming using the plot, what type of correlation is present between the variables? (3 marks)

Solution: There is a negative correlation between the variables, as the gradient of the line is negative. This matches the downward slope of points on the plot. [3 marks]

(iii) Interpret in context the meaning of the value for the gradient of your regression line, and the approximate meaning of the value for the y-intercept. (4 marks)

Solution: The gradient is -3.24, which means that each month the pressure in the tyres drops by 3.24 kPa. [2 marks]

The y-intercept is 261.55, which is approximately the initial tyre pressure. [2 marks]

(b) Is a linear model suitable for this situation? Explain why. (3 marks)

Solution: The tyre loses pressure at a roughly constant rate over time, which means a linear model is suitable. The graph also shows linear behaviour. [3 marks]

Total marks for question 4: [20]